FRC SENSOR CAN — REVISION A PROTOTYPE

Four industrial / logic inputs. Protected battery and regulated 5 V sensor power.
STM32F042 + TCAN3413. Digital status and edge counts over a custom team CAN protocol.
Not a CTRE Phoenix–compatible device. Inspect and bench–test prototypes before robot use.

Battery entry and internal supplies

File: power.kicad_sch

Protected sensor power rails

File: sensor_power.kicad_sch

Controller, clock, debug and status

File: controller.kicad_sch

CAN physical interface and termination

File: can.kicad_sch

Input 1: industrial / logic selection

File: input1.kicad_sch

Input 2: industrial / logic selection

File: input2.kicad_sch

Input 3: industrial / logic selection

File: input3.kicad_sch

Input 4: industrial / logic selection

File: input4.kicad_sch

Original design — custom WPILib CAN protocol

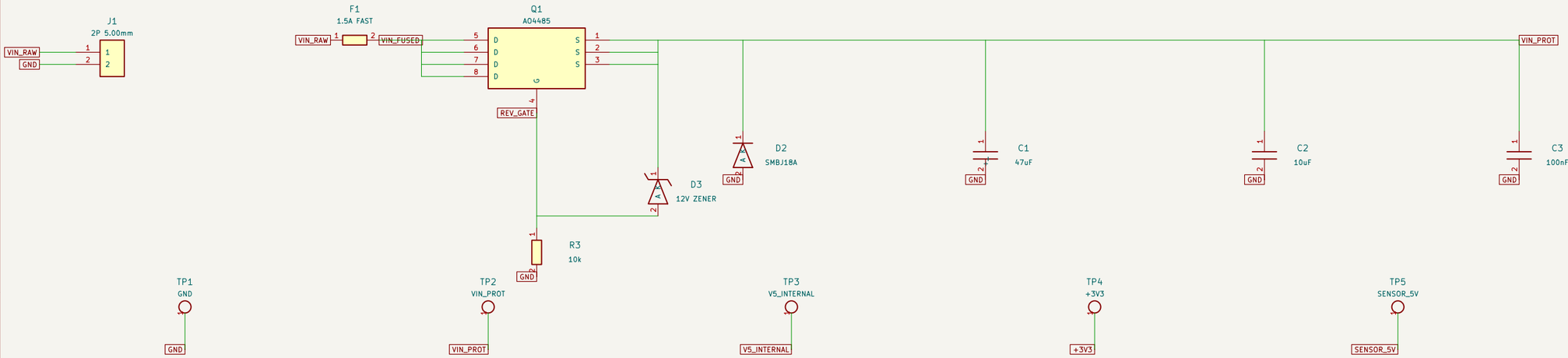
Sheet: /
File: FRC_Sensor_CAN.kicad_sch

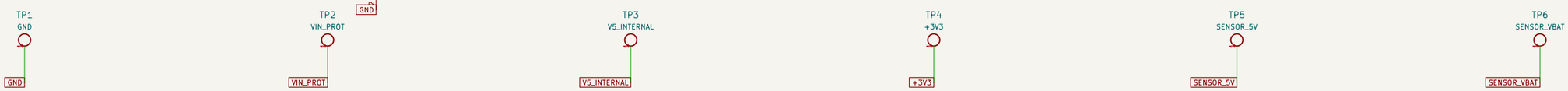
Title: FRC Sensor CAN

Size: A3	Date: 2026-09-07	**Rev: A PROTOTYPE**
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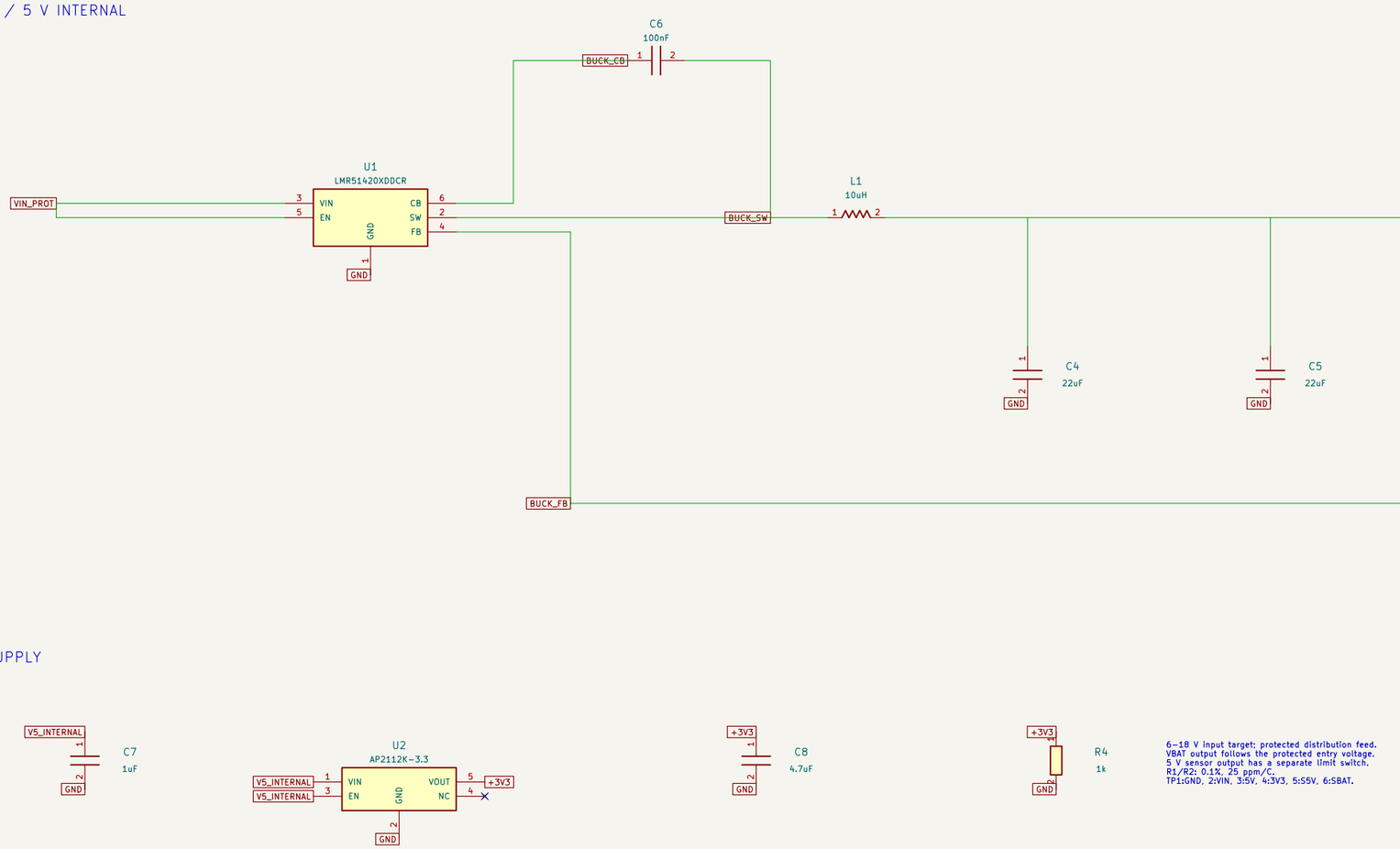
Battery entry and internal supplies

PROTECTED BATTERY ENTRY





500 kHz BUCK / 5 V INTERNAL



3.3 V LOGIC SUPPLY

6–18 V input target; protected distribution feed.
VBAT output follows the protected entry voltage.
5 V sensor output has a separate limit switch.
R1/R2: 0.1%, 25 ppm/C.
TP1:GND, 2:VIN, 3:5V, 4:3V3, 5:S5V, 6:SBAT.

Prototype: bench validation required before robot use

Original sensor interface design

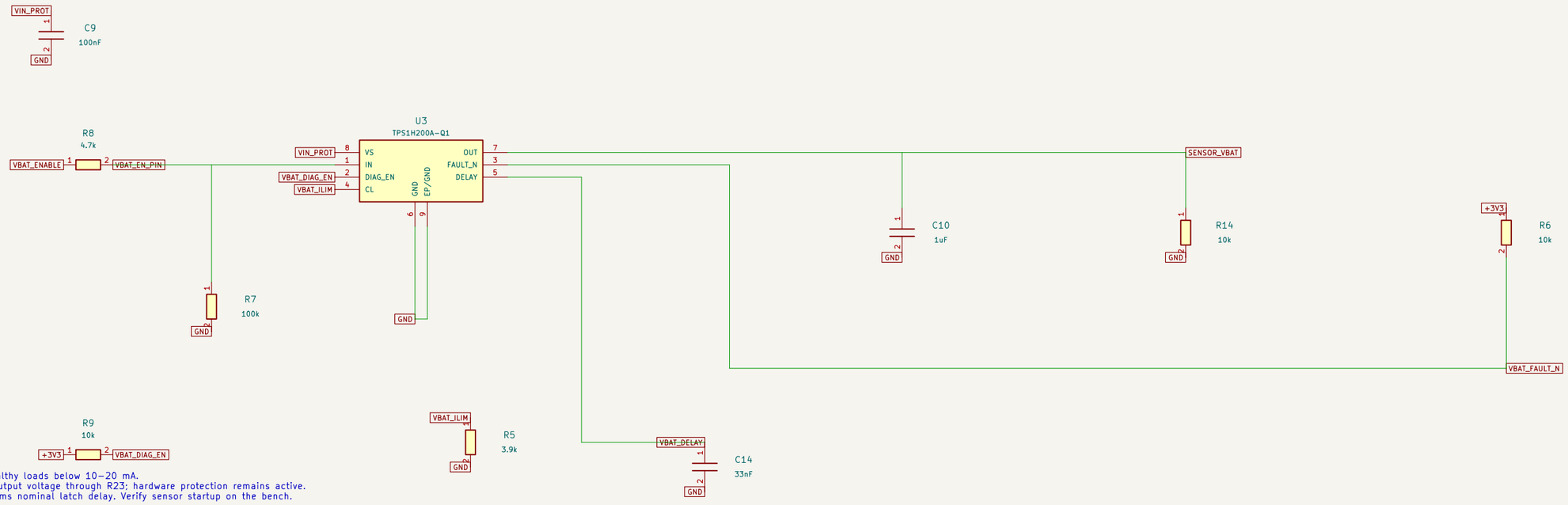
Sheet: /Battery entry and internal supplies/
File: power.kicad_sch

Title: FRC Sensor CAN — Battery entry and internal supplies

Size: A3	Date: 2026-09-07	**Rev: A PROTOTYPE**
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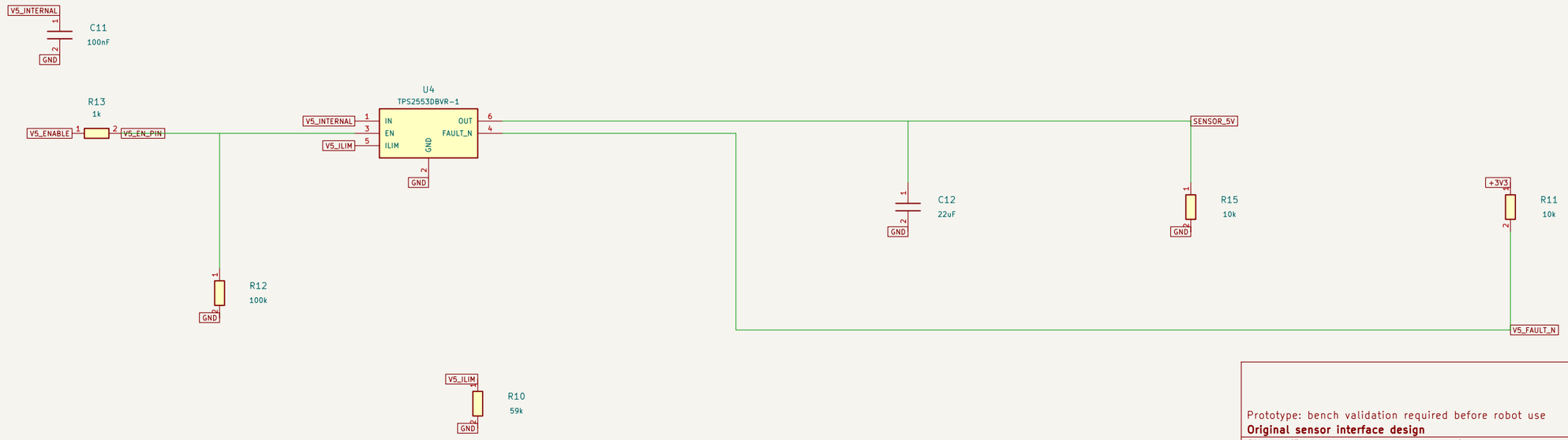
Protected sensor power rails

PROTECTED BATTERY PASSTHROUGH



U3 FAULT also asserts for healthy loads below 10–20 mA.
Firmware monitors switched output voltage through R23; hardware protection remains active.
C14 = 33 nF gives about 11 ms nominal latch delay. Verify sensor startup on the bench.

REGULATED 5 V SENSOR OUTPUT



Usable targets: 300 mA total on each rail, shared across all ports.
Both enable pins have hardware pulldowns. Firmware latches overload faults.
TPS2553 -1 is the latching version. Clear only after inspecting sensor wiring.

Prototype: bench validation required before robot use
Original sensor interface design

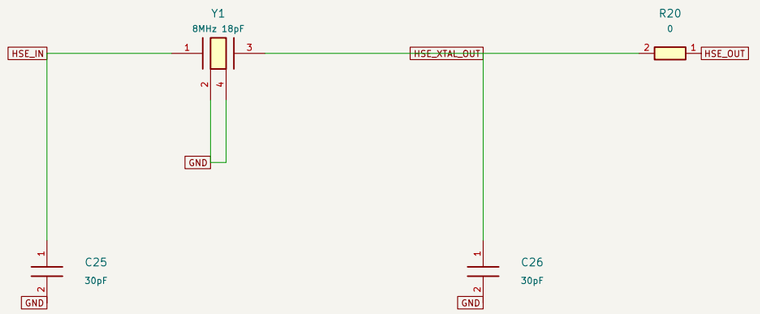
Sheet: /Protected sensor power rails/
File: sensor_power.kicad_sch

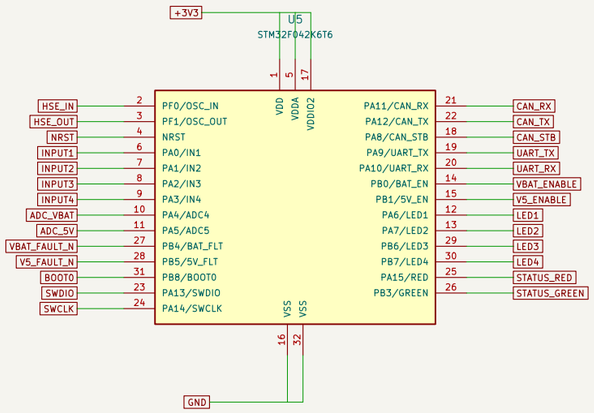
Title: FRC Sensor CAN – Protected sensor power rails

Size: A3	Date: 2026-09-07	**Rev: A PROTOTYPE**
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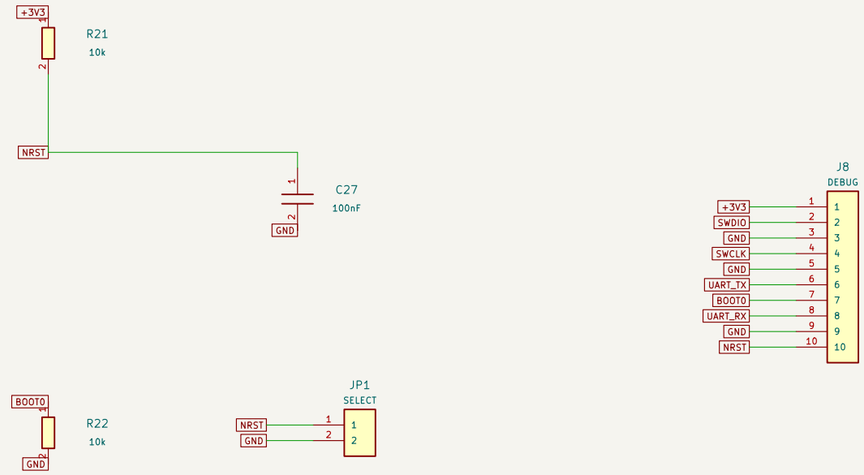
Controller, clock, debug and status

8 MHz CRYSTAL / PLL x6

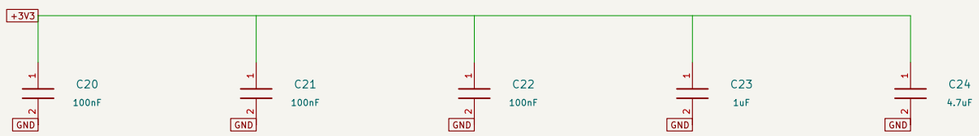




RESET / BOOT / SERVICE



LOCAL DECOUPLING



RAIL MONITORING



J8 is a CUSTOM 2.54 mm service header.
Pin 1 is a 3.3 V reference OUTPUT only.
Use a 3.3 V SWD / UART adapter and shared GND.
BOOT0 normally low; JP1 is momentary reset.
CAN PA11/PA12 use AF4. LEDs are on CAN sheet.

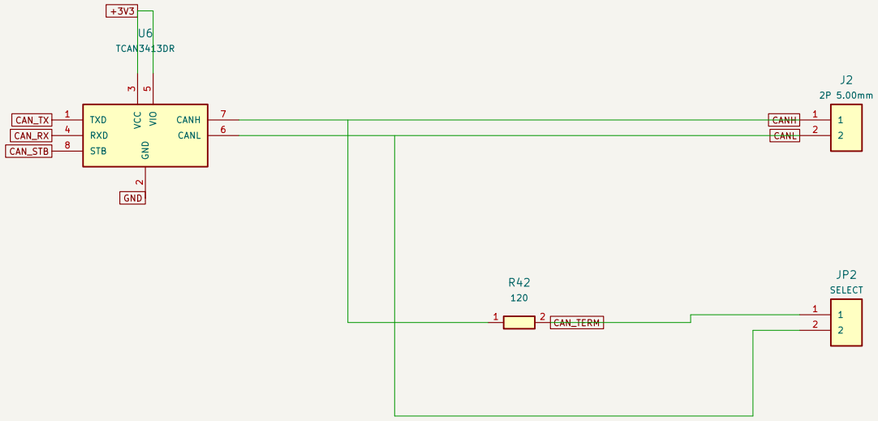
Prototype: bench validation required before robot use
Original sensor interface design

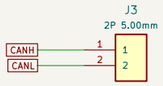
Sheet: /Controller, clock, debug and status/
File: controller.kicad_sch

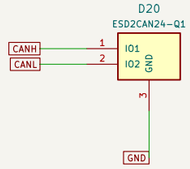
Title: FRC Sensor CAN — Controller, clock, debug and status

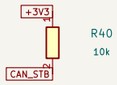
Size: A3	Date: 2026-09-07	**Rev: A PROTOTYPE**
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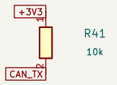
CAN physical interface and termination



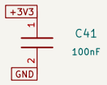








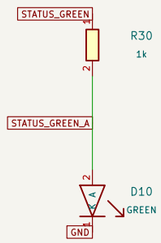


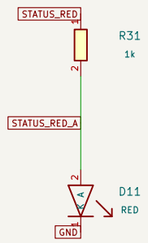




1 Mbit/s. Fit JP2 only at a physical CAN trunk end.
TCAN3413 uses 3.3 V for both VCC and VIO.
STB pull-up keeps the bus silent during MCU reset.
CAN ground is shared through the power connection.

STATUS AND FOUR INPUT INDICATORS













Prototype: bench validation required before robot use

Original sensor interface design

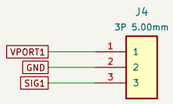
Sheet: /CAN physical interface and termination/
File: can.kicad_sch

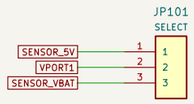
Title: FRC Sensor CAN — CAN physical interface and termination

Size: A3	Date: 2026-09-07	**Rev: A PROTOTYPE**
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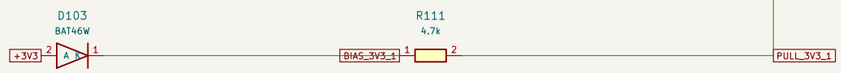
Input 1: industrial / logic selection

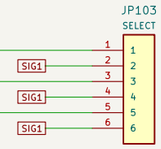
PORT POWER AND BIAS







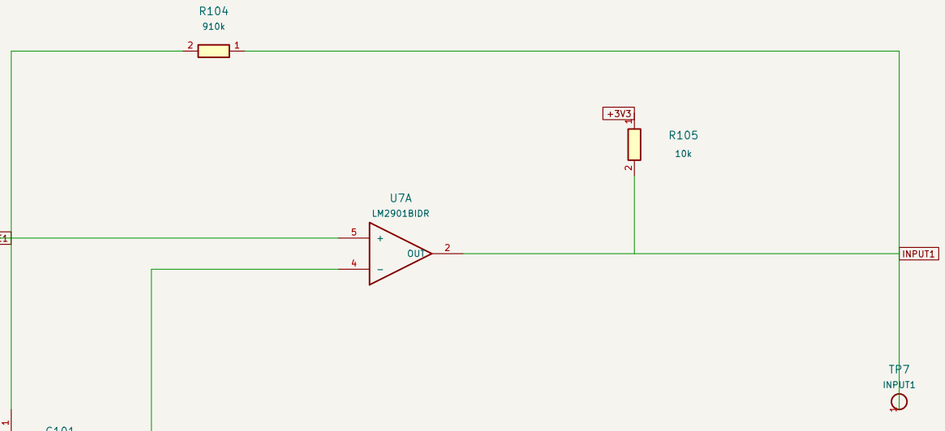


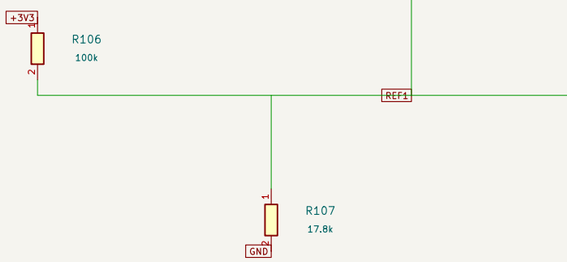




One shunt per selector. Bias diodes block signal backfeed into the supply rails.
NPN: port pullup. PNP: pulldown. Push–pull: bias open. Each bias branch is current limited.

PROTECTED SIGNAL AND HYSTERESIS







Threshold jumper open: LOW <=4V, HIGH >=7V.
Jumper fitted: LOW <=0.8V, HIGH >=2.4V.
0–30V signal target. 1nF input filter; digital debounce in firmware.
Do not apply external power to VPORT.

Prototype: bench validation required before robot use
Original sensor interface design

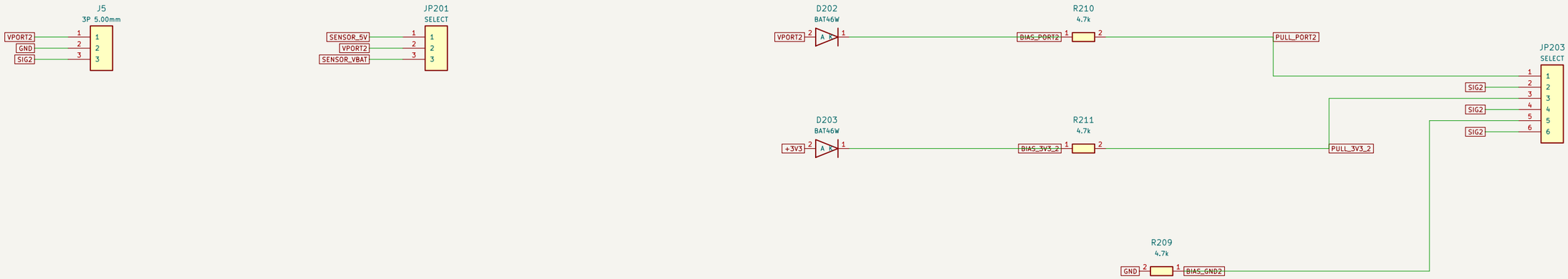
Sheet: /Input 1: industrial / logic selection/
File: input1.kicad_sch

Title: FRC Sensor CAN – Input 1: industrial / logic selection

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Input 2: industrial / logic selection

PORT POWER AND BIAS



One shunt per selector. Bias diodes block signal backfeed into the supply rails.
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PROTECTED SIGNAL AND HYSTERESIS

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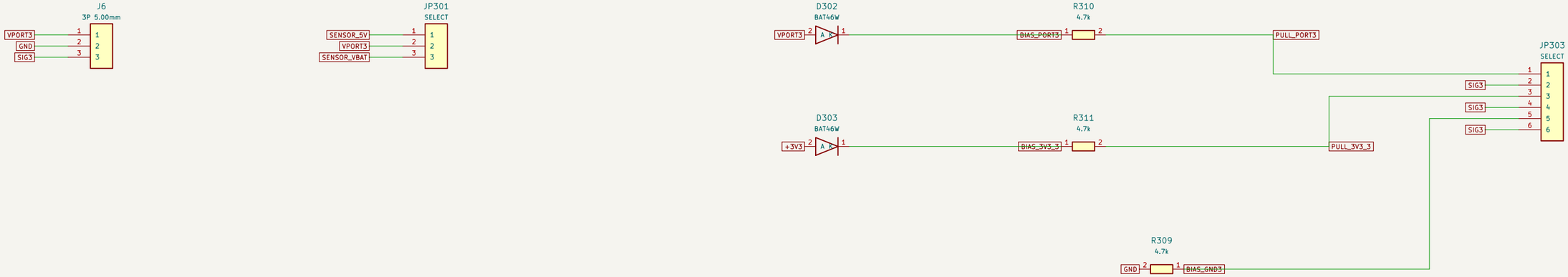
Sheet: /Input 2: industrial / logic selection/
File: input2.kicad_sch

Title: FRC Sensor CAN – Input 2: industrial / logic selection

Size: A3	Date: 2026-09-07	**Rev: A PROTOTYPE**
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Input 3: industrial / logic selection

PORT POWER AND BIAS



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Original sensor interface design

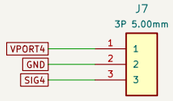
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File: input3.kicad_sch

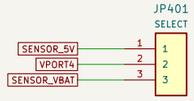
Title: FRC Sensor CAN – Input 3: industrial / logic selection

Size: A3	Date: 2026-09-07	**Rev: A PROTOTYPE**
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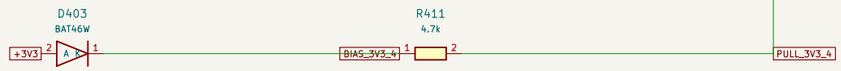
Input 4: industrial / logic selection

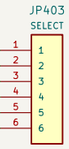
PORT POWER AND BIAS

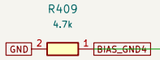






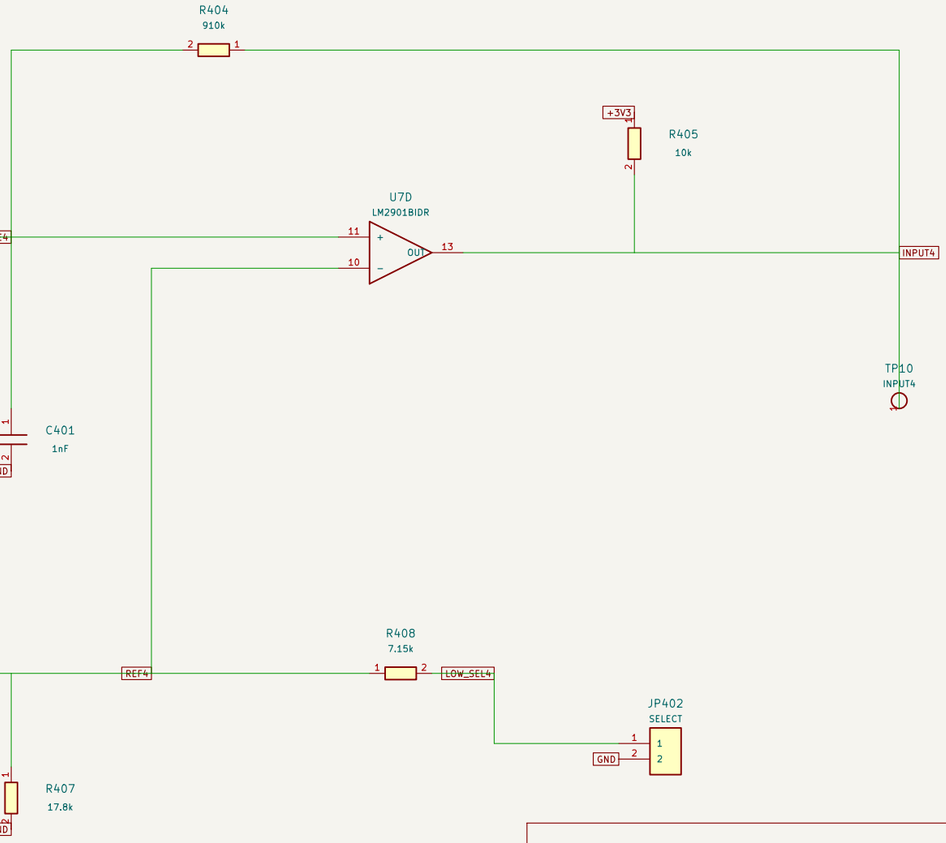






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Do not apply external power to VPORT.

Prototype: bench validation required before robot use
Original sensor interface design

Sheet: /Input 4: industrial / logic selection/
File: input4.kicad_sch

Title: FRC Sensor CAN – Input 4: industrial / logic selection

Size: A3	Date: 2026-09-07	**Rev: A PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 9/9